# Orynt

- Orynt: every payment, under control
- Authorization, single-send safety and verifiable receipts for Lightning payouts
- Dakar Bitcoin Hack 2026 · working title · regtest only
- Not a wallet. Not a launched product.



# The problem (a hypothesis)

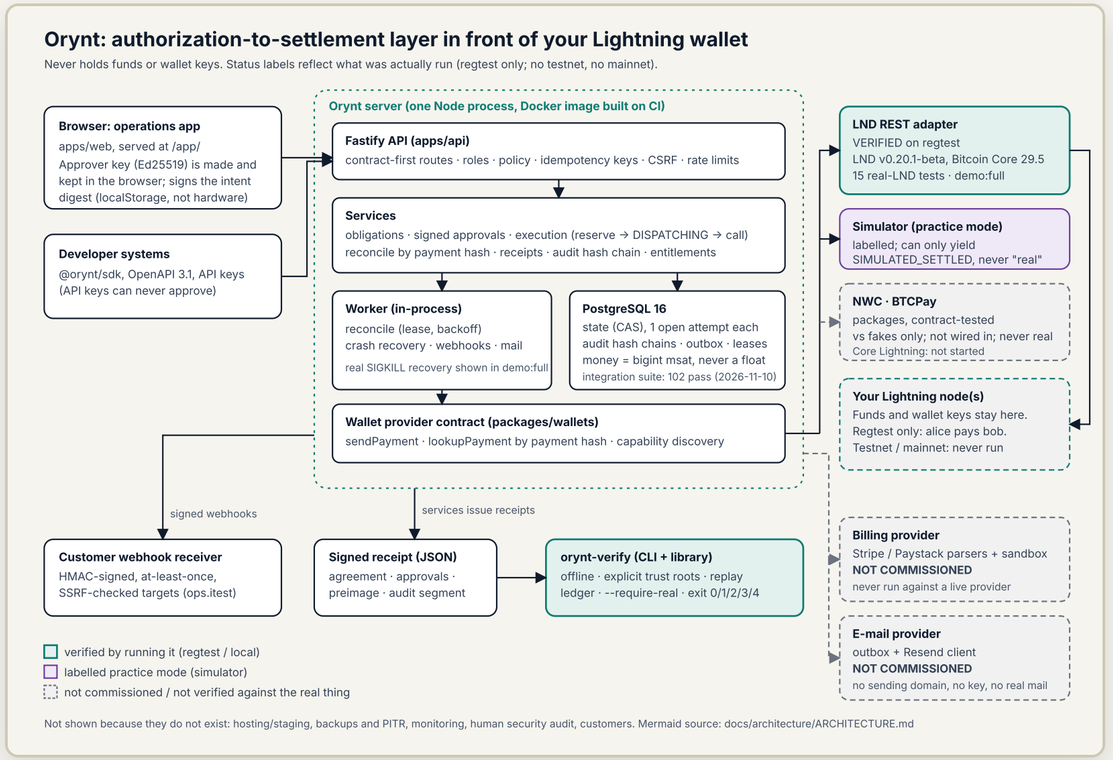
- Bitcoin teams pay contractors, contributors and suppliers in sats
- Today: obligation in a sheet or chat, approval in a message, payment in a wallet screen, proof as a screenshot
- When the wallet's answer is lost mid-payment: retry and risk paying twice, or stall
- Hypothesis from desk research. No interviews, pilots or users yet

# Who it is for, and what is not claimed

- Primary users (hypothesis): the payout operator and approvers at community groups, grant programmes, open-source projects and small Bitcoin businesses
- Cross-border payouts with few people to double-check make mistakes costly
- Not claimed: market size, reach, users, traction
- Plan: 20 structured interviews, at least 4 in Africa, with a written stop criterion

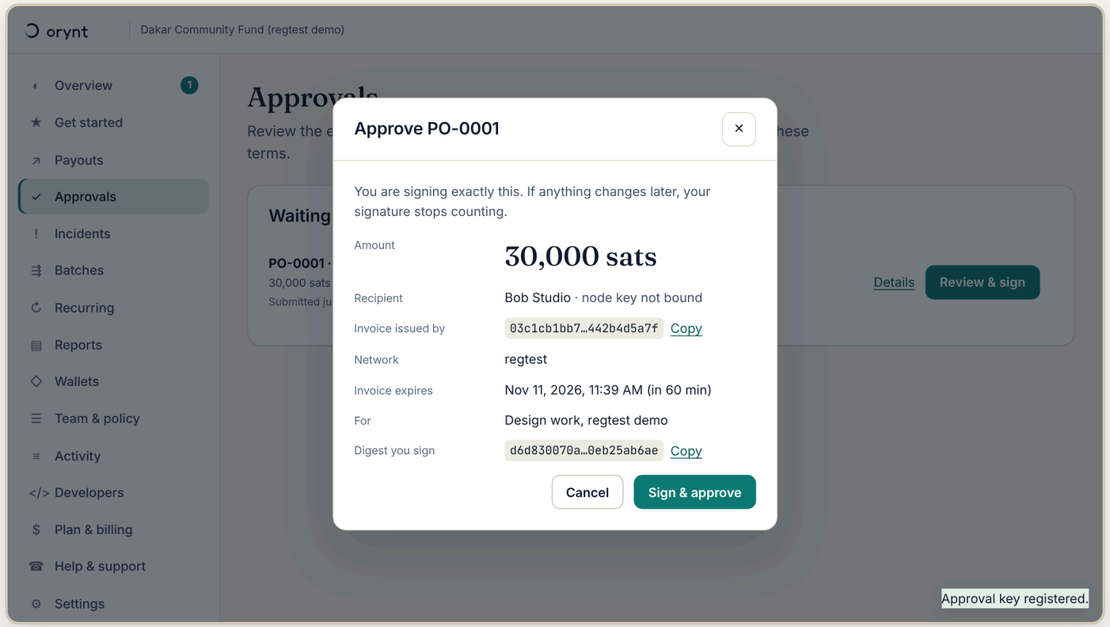
# What Orynt is

- A control layer in front of a wallet you already run; it holds no funds or keys
- Integrates with LND over REST (verified on regtest); labelled simulator for practice
- Obligation, approval, execution, truthful state, evidence
- Billing and e-mail providers are drawn dashed: not commissioned



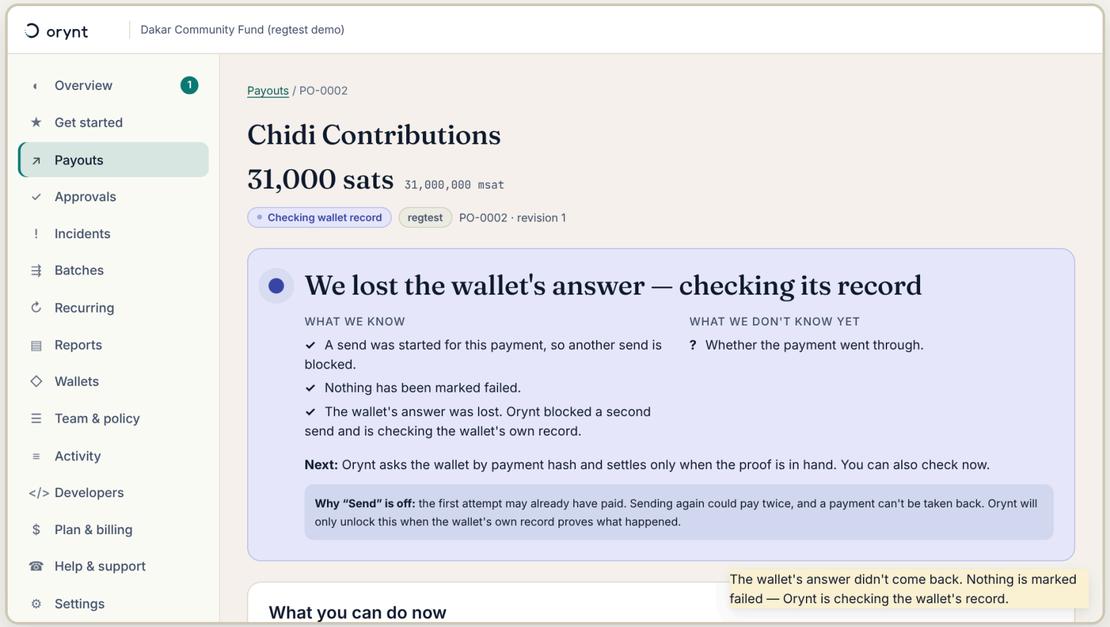
# Authorization: signed approvals

- Default policy: 1 approver up to 100,000 sats, 2 above; the creator cannot approve
- Approvers sign the exact payment: revision, amount, hash, payee key, expiry, policy, wallet
- Edit anything and earlier approvals are void; roles are re-checked at execution
- Enforced by Orynt, not by the wallet: anyone holding the wallet credential bypasses it



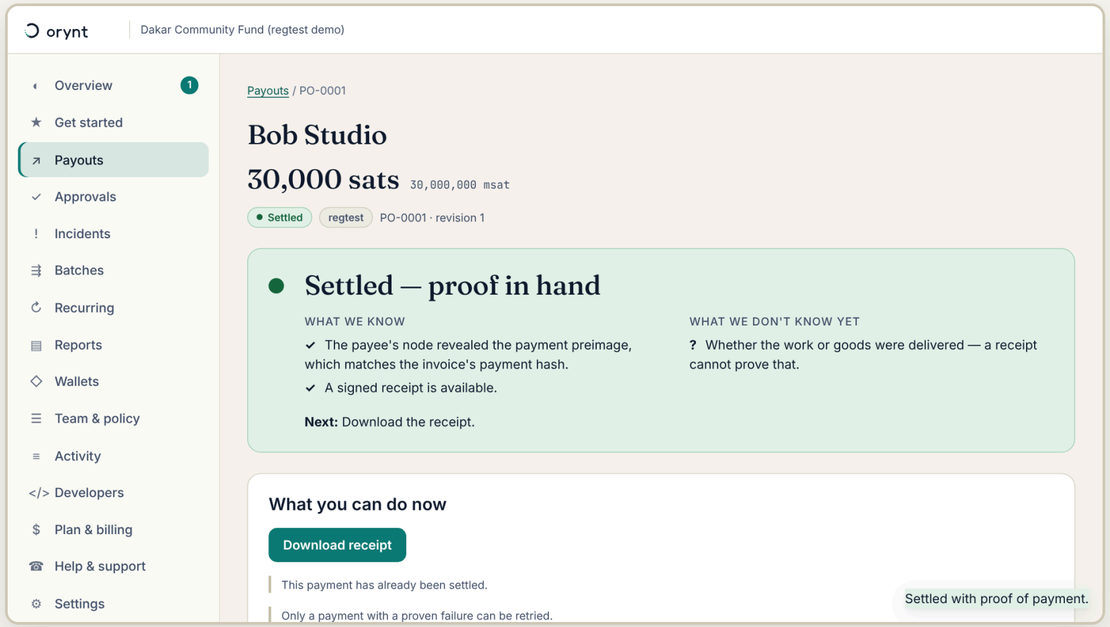
# Exactly one send, and the UNKNOWN state

- Reserve in PostgreSQL, mark DISPATCHING, then call the wallet
- Lost response means UNKNOWN, never FAILED; no path from UNKNOWN back to a send
- A second send is refused (HTTP 409) until the wallet's own record is read by payment hash
- Real LND: 10 concurrent executes, 1 payment



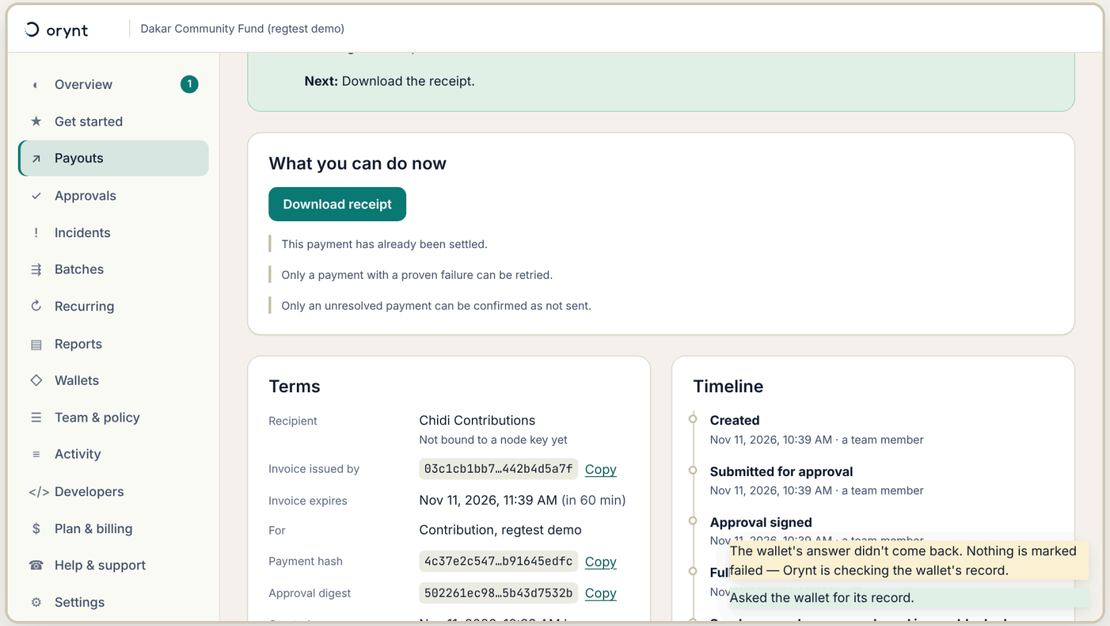
# Evidence you can check without us

- Receipt = agreement + signed approvals + preimage + audit segment + issuer signature + what it cannot prove
- orynt-verify runs offline: explicit trust root, replay ledger, --require-real
- Real receipts: VERIFIED (exit 0); one changed character: REJECTED (exit 1); no trust root: exit 3
- A simulator receipt is refused under --require-real



# What I actually ran (real regtest)

- pnpm demo:full against Bitcoin Core 29.5 + two LND v0.20.1-beta nodes: all assertions pass (2026-11-11)
- Real payment settled once; lost response (injected) reconciled; API SIGKILLed mid-payment and recovered
- 15 real-LND tests pass; 102 integration, 51 unit, 138 browser checks; 6 CI jobs green
- Regtest only. Never run on testnet or mainnet. Only LND verified



# Security by design, and what is missing

- Done: non-custodial; least-privilege roles; wallet permissions discovered; secrets encrypted at rest; validated inputs; tenant isolation; SSRF guards
- Done: TOTP MFA step-up (opt-in per user); hash-chained audit log
- Missing: human audit or pen test (the review was by AI); hardware-backed approval keys; SSO
- Missing: managed-node check against fake-LND "REAL" receipts; mature BOLT11 library

# How it is built

- TypeScript monorepo; Fastify + PostgreSQL; Zod contract generates OpenAPI 3.1 and a typed SDK
- One wallet interface: LND REST adapter and a labelled simulator; NWC and BTCPay adapters contract-tested against local fakes only, not wired in
- Standalone verifier; deterministic fault injection; operations app and 18-page site
- CI on GitHub: lint, typecheck, tests, real-Lightning job, browser E2E, container build

# Impact, and how we will find out

- Hypothesis: one place for obligation, approval, send and proof
- To measure: onboarding completion, time to first authorised payout, exception time-to-resolution, duplicates blocked per 100 payouts (with n)
- No users, revenue, partners or reach estimate exist today
- Kill criterion: if interviews show nobody has lost or doubled a payout, stop

# Not verified, next, and the ask

- Not verified: testnet, mainnet, other wallets, human audit, legal review, hosting, real e-mail and billing, any customer
- Next (plans): interviews, verified NWC and BTCPay adapters, hardware-backed approval keys, mobile-first approvals
- Ask: tell me whether payout control is a real problem; introductions to teams that pay contributors
- Working title; trademark unresolved. Code and evidence: see the Devpost page